Sampling & Estimation

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Solve the equation $\ln(3x+1) - 2\ln 2 = \ln(x-1)$.

Using the power law of logarithms on the second term: $2\ln 2 = \ln(2^2) = \ln 4$. Substitute this back into the equation: $\ln(3x+1) - \ln 4 = \ln(x-1)$. Apply the quotient rule for logarithms, $\ln A - \ln B = \ln\left(\frac{A}{B}\right)$: $\ln\left(\frac{3x+1}{4}\right) = \ln(x-1)$. Since the natural logarithm function is one-to-one, we equate the arguments: $\frac{3x+1}{4} = x-1$. Multiply both sides by 4: $3x+1 = 4(x-1)$, $3x+1 = 4x-4$. Rearranging gives.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The masses, x grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.

(a) Sample mean: $\bar{x} = \frac{5090}{50} = 101.8$ g Unbiased sample variance: $s^2 = \frac{1}{50-1}(518380 - 50(101.8)^2) = \frac{1}{49}(518380 - 518162) = \frac{218}{49} \approx 4.449$ g^2 (b) Standard error: $\text{SE} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{4.449}{50}} \approx 0.2983$.

$\bar{x} \pm 1.96(0.2983) = 101.8 \pm 0.585 = (101.2,\ 102.4)$ g (1 d. p.)

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.

$142 \pm 2.576 \cdot \frac{20}{\sqrt{100}} = 142 \pm 5.15 = (136.85,\ 147.15)$. "99% confidence" means: if this process were repeated many times, 99% of the intervals produced would contain the true μ.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine whether the lines $\mathbf{r}_1 = (1; 2; -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

1. Check parallelism: Direction vectors are $\mathbf{d}_1 = (2; 1; 3)$ and $\mathbf{d}_2 = (-1; 2; 1)$. Since $\frac{2}{-1} eq \frac{1}{2}$, they are not scalar multiples, so the lines are not parallel. 2. Equate components: $1 + 2\lambda = -\mu \Rightarrow 2\lambda + \mu = -1 \quad (1)$ $2 + \lambda = 5 + 2\mu \Rightarrow \lambda - 2\mu = 3 \quad (2)$ $-1 + 3\lambda = 4 + \mu \Rightarrow 3\lambda - \mu = 5 \quad (3)$ Add (1) and (3): $5\lambda = 4 \Rightarrow \lambda = \frac{4}{5}$. From.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.

Rewrite with negative fractional index and factor out 4: $(4-x)^{-1/2} = 4^{-1/2}(1-\frac{x}{4})^{-1/2} = \frac{1}{2}(1-\frac{x}{4})^{-1/2}$ Apply $(1+y)^n = 1 + ny + \frac{n(n-1)}{2}y^2 + \cdots$ with $n = -\frac{1}{2}$ and $y = -\frac{x}{4}$: $(1-\frac{x}{4})^{-1/2} = 1 + (-\frac{1}{2})(-\frac{x}{4}) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2}(-\frac{x}{4})^2 + \cdots$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

Let $u = 2x + 1$, so $du = 2\,dx \Rightarrow dx = \frac{1}{2}\,du$, and $x = \frac{u-1}{2}$. Change limits: $x = 0 \Rightarrow u = 1, \quad x = 4 \Rightarrow u = 9$ Transform the integral: $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx = \int_1^9 \frac{\frac{u-1}{2}}{\sqrt{u}} \cdot \frac{1}{2}\,du = \frac{1}{4}\int_1^9 (u^{1/2} - u^{-1/2})\,du$ $= \frac{1}{4}\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^9 = \frac{1}{4}\left(\left(\frac{2}{3}(27) - 2(3)\right) - \left(\frac{2}{3}(1) - 2(1)\right)\right)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

(a) Under H_0, $X \sim \text{Po}(6)$. Calculate cumulative probabilities: $P(X = 0) = e^{-6} \approx 0.00248$ $P(X \le 1) = e^{-6}(1 + 6) = 7e^{-6} \approx 0.01735$ $P(X \le 2) = e^{-6}(1 + 6 + 18) = 25e^{-6} \approx 0.06197$ Since $P(X \le 1)0.05$, the critical region is $X \le 1$ (i.e. $X = 0$ or 1). (b) Actual significance level = $P(X \le 1 \mid \lambda = 6) = 7e^{-6} \approx 0.0174$ (or 1.74%). (c) Type II error occurs when we fail to reject H_0 when $\lambda = 3$. We fail to reject when.

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9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

Substitute the double angle identity $\cos 2x = 1 - 2\sin^2 x$: $2(1 - 2\sin^2 x) = 1 + \sin x$
$2 - 4\sin^2 x = 1 + \sin x \Rightarrow 4\sin^2 x + \sin x - 1 = 0$ Apply quadratic formula for $\sin x$:
$\sin x = \frac{-1 \pm \sqrt{1 - 4(4)(-1)}}{8} = \frac{-1 \pm \sqrt{17}}{8}$ Case 1: $\sin x = \frac{-1 + \sqrt{17}}{8} \approx 0.3904$:
$x = 0.401$ rad, $\pi - 0.401 = 2.74$ rad Case 2: $\sin x = \frac{-1 - \sqrt{17}}{8} \approx -0.6404$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

Differentiate implicitly with respect to x: $3x^2 - 2\left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$
$3x^2 - 2y + (2y - 2x)\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2y - 3x^2}{2y - 2x}$ A vertical tangent occurs when the denominator is zero (and numerator non-zero): $2y - 2x = 0 \Rightarrow y = x$ Substitute $y = x$ into the original curve equation:
$x^3 - 2x(x) + x^2 = 4 \Rightarrow x^3 - x^2 - 4 = 0$ Test integer factors: $x = 2$ gives $8 - 4 - 4 = 0$. Factoring gives $(x - 2)(x^2 + x + 2) = 0$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

(a) In a 2-week period, the parameter scales to $\lambda = 1.6 \times 2 = 3.2$:
$P(X \ge 3) = 1 - P(X \le 2) = 1 - e^{-3.2}\left(1 + 3.2 + \frac{3.2^2}{2}\right) = 1 - 9.32e^{-3.2} \approx 1 - 0.3799 = 0.620$ (3 s. f.) (b) For 1 week, $\lambda = 1.6$. Use conditional probability:
$P(X = 2 \mid X \ge 1) = \frac{P(X=2)}{P(X \ge 1)} = \frac{\frac{e^{-1.6}(1.6)^2}{2}}{1 - e^{-1.6}} = \frac{1.28e^{-1.6}}{1 - e^{-1.6}} \approx \frac{0.2584}{0.7981} \approx 0.324$ (3 s. f.)

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

On an Argand diagram, the point P represents the complex number z such that $|z - (2 + 3i)| \le 2$. Find the greatest possible value of $|z|$ for points in this region.

The locus $|z - (2 + 3i)| \le 2$ represents a circular disc of radius $R = 2$ with center at $C(2, 3)$. The distance from the origin $O(0, 0)$ to the center C is: $|C| = \sqrt{2^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13} \approx 3.6055$ The point on the disc farthest from the origin lies on the line passing through O and C, on the far boundary of the circle:
$|z|_{\max} = |C| + R = \sqrt{13} + 2 \approx 3.606 + 2 = 5.61$ (3 s. f.)

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95 - 1.10) = 1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

Integrate exactly: $\int_1^3 \frac{4}{2x+1}\,dx = [2\ln(2x+1)]_1^3 = 2\ln 7 - 2\ln 3 = 2\ln\left(\frac{7}{3}\right) \approx 1.6946\ldots$ The exact value is $2\ln\left(\frac{7}{3}\right)$. The premature rounding gives 1.70, yielding an absolute error of: $|1.70 - 1.6946| \approx 0.0054$ On Cambridge Paper 3, this premature rounding costs the final accuracy mark (A1) because the 3 s.f. answer is 1.69, not 1.70.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.

(a) Let $g(x) = (x+1)^{1/3}$. Differentiate: $g'(x) = \frac{1}{3}(x+1)^{-2/3} = \frac{1}{3(x+1)^{2/3}}$ Near the root $x \approx 1.3$: $g'(1.3) = \frac{1}{3(2.3)^{2/3}} \approx \frac{1}{3(1.742)} \approx 0.191$ Since \(|g'(x)| $x_1 = 1.3$ $x_2 = (1.3+1)^{1/3} = (2.3)^{1/3} \approx 1.3200$ $x_3 = (1.3200+1)^{1/3} = (2.3200)^{1/3} \approx 1.3238$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 15.01 Hypothesis Testing

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

Take natural logarithms of $y = ax^b$: $\ln y = \ln a + b\ln x$ This is a linear equation $Y = mX + c$ with $Y = \ln y$, $X = \ln x$, gradient $m = b$, and intercept $c = \ln a$. Gradient: $b = \frac{6.80 - 1.20}{4 - 0} = \frac{5.60}{4} = 1.40$ Vertical intercept gives $\ln a = 1.20$: $a = e^{1.20} \approx 3.32$ Therefore, $a = 3.32$ and $b = 1.40$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Exponentials & Logarithms

$f(x) = cx^2$ for $0 \le x \le 3$, zero otherwise. Find c, $E(X)$, and $P(1 \le X \le 2)$.

$\int_0^3 cx^2\,dx = 9c = 1 \Rightarrow c = \frac{1}{9}$. $E(X) = \frac{1}{9}\int_0^3 x^3\,dx = \frac{1}{9}\cdot\frac{81}{4} = \frac{9}{4}$. $P(1 \le X \le 2) = \frac{1}{9}\left[\frac{x^3}{3}\right]_1^2 = \frac{1}{27}(8-1) = \frac{7}{27}$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Advanced Trigonometry

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

$H_0 : \mu = 80, H_1 : \mu < 80$. $Z = \frac{77.5 - 80}{5/4} = -\frac{2.5}{1.25} = -2$. Critical value at 1%: -2.326. Since $-2 > -2.326$, do not reject H_0. Insufficient evidence at 1% that mean breaking strength is below 80 kN.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Differentiation

Lines ℓ_1 and ℓ_2 have direction vectors $\mathbf{d_1} = (2; 1; -1)$ and $\mathbf{d_2} = (1; -1; 3)$. Find the acute angle between them.

$\mathbf{d_1} \cdot \mathbf{d_2} = 2 - 1 - 3 = -2$. $|d_1| = 6$, $|\mathbf{d_2}| = \sqrt{11}$. $\cos\theta = \frac{2}{\sqrt{66}} \Rightarrow \theta = \arccos\left(\frac{2}{\sqrt{66}}\right) \approx 75.7°$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Integration

Express $\frac{3x+1}{(x+1)(x-2)}$ in partial fractions.

$\frac{A}{x+1} + \frac{B}{x-2}$. Cover-up: let $x = 2$: $B = \frac{7}{3}$. Let $x = -1$: $A = \frac{-2}{-3} = \frac{2}{3}$. Answer: $\frac{2/3}{x+1} + \frac{7/3}{x-2}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Sequences & Series

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

(a) $P(X = 5) = \frac{e^{-3} \cdot 243}{120} \approx 0.1008$. (b) $\lambda = 1.5$: $P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-1.5} - 1.5e^{-1.5} = 1 - 2.5e^{-1.5} \approx 0.4422$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the unbiased estimator of the population variance σ^2 calculated from a random sample of size n?

a. $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ ✓

b. $s^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum(x_i - \bar{x})^2$

d. $s^2 = \sum(x_i - \bar{x})^2$

Dividing by $n - 1$ instead of n removes bias, making $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ the unbiased estimator of σ^2.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random sample of 10 observations has $\sum x = 80$ and $\sum x^2 = 724$. Calculate the unbiased estimate of the population variance.

a. 9.33 ✓

b. 8.40

c. 10.50

d. 7.20

$s^2 = \frac{1}{n-1}\left[\sum x^2 - \frac{(\sum x)^2}{n}\right] = \frac{1}{9}\left[724 - \frac{6400}{10}\right] = \frac{1}{9}[724 - 640] = \frac{84}{9} \approx 9.33.$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the biased sample variance is $v = 18$ from a sample of size $n = 25$, find the unbiased estimate s^2.

a. 18.75 ✓

b. 17.28

c. 18.00

d. 19.50

$s^2 = \frac{n}{n-1}v = \frac{25}{24}(18) = 18.75.$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the Central Limit Theorem for the distribution of the sample mean $\bar{X}$.

a. **For large n, $\bar{X}$ is approximately normal $N(\mu, \sigma^2/n)$ regardless of the population distribution** ✓

b. For any n, $\bar{X}$ is strictly normal

c. For large n, the population becomes normal

d. $\bar{X}$ has mean $n\mu$ and variance $n\sigma^2$

The CLT states that the sample mean $\bar{X}$ of a random sample from ANY distribution with mean μ and variance σ^2 is approximately normally distributed $N(\mu, \sigma^2/n)$ as $n \to \infty$ (typically $n \geq 30$).

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A population has mean 100 and standard deviation 15. A sample of size $n = 36$ is taken. What is the standard error of the sample mean?

a. 2.5 ✓

b. 15

c. 4.17

d. 0.417

Standard error $\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{36}} = \frac{15}{6} = 2.5.$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Using the distribution from the previous question ($\mu = 100$, SE $= 2.5$), find $P(\bar{X} > 105)$.

a. 0.0228 ✓

b. 0.0500

c. 0.0456

d. 0.0100

$z = \frac{105 - 100}{2.5} = 2.0$. $P(\bar{X} > 105) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A sample of size $n = 64$ from a normal population with known $\sigma = 8$ has sample mean $\bar{x} = 52.0$. Find the 95% confidence interval for μ.

a. [50.04, 53.96] ✓

b. [49.50, 54.50]

c. [50.36, 53.64]

d. [51.00, 53.00]

Margin of error $= z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{8}{8} = 1.96$. Interval $= 52.0 \pm 1.96 = [50.04, 53.96]$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the critical value z used to construct a 99% confidence interval for the population mean?

a. 2.576 ✓

b. 1.960

c. 1.645

d. 2.326

For a 99% two-tailed confidence interval, $1 - \alpha = 0.99 \Rightarrow \alpha/2 = 0.005$. $\Phi(2.576) = 0.995$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How large must a sample be so that the width of a 95% confidence interval for μ is at most 2 units, given $\sigma = 10$?

a. **At least 385** ✓

b. At least 100

c. At least 196

d. At least 250

Width $= 2z\frac{\sigma}{\sqrt{n}} \le 2 \Rightarrow z\frac{\sigma}{\sqrt{n}} \le 1 \Rightarrow 1.96 \times \frac{10}{\sqrt{n}} \le 1 \Rightarrow \sqrt{n} \ge 19.6 \Rightarrow n \ge 19.6^2 = 384.16$. Least integer is 385.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

When σ^2 is unknown but the sample size $n = 100$ is large, what is substituted for σ^2 in calculating the confidence interval?

a. **The unbiased sample variance estimate s^2** ✓

b. The sample range divided by 4

c. The median absolute deviation

d. A default value of 1.0

By the Central Limit Theorem and Slutsky's theorem, when n is large, the unknown population variance σ^2 is replaced by the unbiased sample variance estimate s^2.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A sample of 100 lightbulbs has mean life $\bar{x} = 1200$ hours and unbiased sample variance $s^2 = 2500$. Find the 90% confidence interval for the true mean life.

a. $[1191.8, 1208.2]$ ✓

b. $[1180.0, 1220.0]$

c. $[1190.2, 1209.8]$

d. $[1195.0, 1205.0]$

$s = \sqrt{2500} = 50$. Standard error $= 50/\sqrt{100} = 5$. For 90% confidence, $z = 1.645$. Margin $= 1.645(5) = 8.225$. Interval is $[1191.8, 1208.2]$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the correct probabilistic interpretation of a 95% confidence interval $[L, U]$ for μ?

a. **In repeated sampling, 95% of such constructed intervals will contain the true parameter μ** ✓

b. There is a 95% probability that μ lies in this specific interval

c. 95% of population values fall between L and U

d. The sample mean falls in $[L, U]$ 95% of the time

The parameter μ is a fixed unknown constant; the interval endpoints are random variables. 95% of intervals generated by this procedure over repeated samples will capture μ.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $\bar{x} = 61$, $z = 1$, $\sigma = 12$, $n = 4$, find the upper confidence limit $\bar{x} + E$.

a. 61

b. **67** ✓

c. 68

d. 55

Applying Sampling & Estimation principles, $E = 6$. Upper limit $= 61 + 6 = 67$. Therefore, the verified correct option is 67.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A required margin of error is $E = 4$, with $z = 1$, $\sigma = 8$. Find the required sample size $n = (z\sigma/E)^2$.

a. 8

b. 5

c. 2

d. **4** ✓

$n = \left(\frac{1 \times 8}{4}\right)^2 = (2)^2 = 4$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A one-tail (upper) hypothesis test has critical value $z = 2.576$ and test statistic $z = 3$. What is the decision?

a. Accept H_1 unconditionally

b. Do not reject H_0

c. **Reject** H_0 ✓

d. Insufficient information

Applying Hypothesis Testing principles, Since $3 > 2.576$, we reject H_0. Therefore, the verified correct option is Reject H_0.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Expand $\cos(x + 45^\circ)$ in terms of $\sin x$ and $\cos x$.

a. $\cos x \cdot \frac{\sqrt{2}}{2} + \sin x \cdot \frac{\sqrt{2}}{2}$

b. $\cos x \cdot \frac{\sqrt{2}}{2} - \sin x \cdot \frac{\sqrt{2}}{2}$ ✓

c. $\sin x \cdot \frac{\sqrt{2}}{2} - \cos x \cdot \frac{\sqrt{2}}{2}$

d. $\cos x \cdot \frac{\sqrt{2}}{2} - \sin x \cdot \frac{\sqrt{2}}{2} + 0$

$\cos(x + 45^\circ) = \cos x \cos 45^\circ - \sin x \sin 45^\circ = \cos x \cdot \frac{\sqrt{2}}{2} - \sin x \cdot \frac{\sqrt{2}}{2}$.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given $\mathbf{a} = (2, -6, -9)$ and $\mathbf{b} = (2, 3, 6)$, find $\cos\theta$, where θ is the angle between them.

a. $\frac{9}{77}$

b. $\frac{68}{77}$

c. $-\frac{34}{9}$

d. $-\frac{68}{77}$ ✓

$\mathbf{a} \cdot \mathbf{b} = -68$, $|\mathbf{a}| = 11$, $|\mathbf{b}| = 7$. $\cos\theta = \frac{-68}{11 \times 7} = -\frac{68}{77}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Differentiation

A curve is defined parametrically by $x = 2t^2$, $y = 3t^3$. Find $\frac{dy}{dx}$ at $t = 1$.

a. $\frac{7}{3}$

b. $\frac{9}{4}$ ✓

c. $-\frac{9}{4}$

d. $\frac{4}{9}$

$\frac{dx}{dt} = 4t$, $\frac{dy}{dt} = 9t^2$. $\frac{dy}{dx} = \frac{9t^2}{4t} = \frac{9t}{4}$. At $t = 1$: $\frac{9}{4}$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Integration

Find the Cartesian equation of the locus $|z - 2| = |z - (-2)|$.

a. **x=0** ✓

b. x=2

c. x=1

d. y=0

$|z - 2| = |z - (-2)|$ is the perpendicular bisector of the segment joining 2 and (-2): $x = \frac{2 + (-2)}{2} = 0$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Sequences & Series

Given $z = r(\cos 90^\circ + i\sin 90^\circ)$, find the principal argument (in degrees, $0 \le \theta < 360$) of z^3.

a. **270** ✓

b. 0

c. 274

d. 90

By De Moivre's theorem, $\arg(z^3) = n\theta = 3 \times 90^\circ = 270^\circ \equiv 270^\circ \pmod{360}$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Vectors

Solve $e^{2x} - 22e^x + 112 = 0$ for x. Give the larger solution in exact form.

a. $\ln 22$

b. $\ln 112$

c. $\ln 14$ ✓

d. $\ln 8$

$(e^x - 8)(e^x - 14) = 0 \Rightarrow e^x = 8$ or $e^x = 14 \Rightarrow x = \ln 8$ or $x = \ln 14$. Larger: $x = \ln 14$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Given $\sin\theta = \frac{3}{5}$, with θ in Quadrant I, find $\tan 2\theta$.

a. $\frac{24}{7}$ ✓

b. $\frac{7}{25}$

c. $-\frac{24}{7}$

d. $\frac{24}{25}$

$\sin 2\theta = \frac{24}{25},\ \cos 2\theta = \frac{7}{25} \Rightarrow \tan 2\theta = \frac{24}{7}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Complex Numbers

Solve $x^2 - 12x + 52 = 0$. Find the root with positive imaginary part.

a. 4+6i

b. -6+4i

c. **6+4i** ✓

d. 6-4i

Using the quadratic formula: roots $= 6 \pm 4i$. Root with positive imaginary part: $6 + 4i$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 1$ and converges to a limit L. Find L.

a. $\frac{5}{2}$

b. $-\frac{3}{2}$

c. 1

d. $\frac{3}{2}$ ✓

At the limit, $L = \frac{1}{3}L + 1 \Rightarrow L(1 - \frac{1}{3}) = 1 \Rightarrow L = \frac{3}{2}$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Numerical Methods

Given $\sin\theta = \frac{5}{13}$, with θ in Quadrant II, find $\sin 2\theta$.

a. $\frac{120}{169}$

b. $\frac{25}{169}$

c. $-\frac{120}{169}$ ✓

d. $\frac{119}{169}$

$\cos\theta = -\frac{12}{13}$. $\sin 2\theta = 2\sin\theta\cos\theta = -\frac{120}{169}$.

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